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RESEARCH ARTICLE

# Species-level drivers of avian centrality within seed-dispersal networks across different levels of organisation

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## Abstract

1. Bird–plant seed-dispersal networks are structural components of ecosystems. The role of bird species in seed-dispersal networks (from less [peripheral] to more connected [central]), determines the interaction patterns and their ecosystem services. These roles may be driven by morphological and functional traits as well as evolutionary, geographical and environmental properties acting at different spatial extents.
2. It is still unknown if such drivers are equally important in determining species centrality at different network levels, from individual local networks to the global meta-network representing interactions across all local networks.
3. Using 308 networks covering five continents and 11 biogeographical regions, we show that at the global meta-network level species' range size was the most important driver of species centrality, with more central species having larger range sizes, which would facilitate the interaction with a higher number of plants and thus the maintenance of seed-dispersal interactions.
4. At the local network level, body mass was the only driver with a significant effect, implying that local factors related to resource availability are more important at this level of network organisation than those related to broad spatial factors such as range sizes. This could also be related to the mismatch between species-level traits, which do not consider intraspecific variation, and the local networks that can depend on such variation.
5. Taken together, our results show that the drivers determining species centrality are relative to the levels of network organisation, suggesting that prediction of species functional roles in seed-dispersal interactions requires combined local and global approaches.

KEYWORDS
macroecology, meta-networks, morphological traits, mutualistic networks, range size

## 1 | INTRODUCTION

Among the greatest anthropogenic effects on ecosystems are the loss of ecological functions and functional integrity, as a result of species local extinctions (Hagen et al., 2012; Loiseau et al., 2020). To mitigate these effects, it is necessary to know what drives species functionality in different ecosystems (Hooper et al., 2005). Potential drivers of such species functionality that have been previously evaluated as independent factors include morphological and functional traits, evolutionary properties, geography and environmental conditions (Barnagaud et al., 2014; Burin et al., 2021). However, the relative importance of these factors is still subject to debate (Coux et al., 2016; Pigot et al., 2016).

Bird seed-dispersal networks are excellent models to evaluate species' ecosystem functions (Schleuning et al., 2015). Indeed, from an ecological perspective, birds are the main seed dispersers, establishing mutualistic interactions with most angiosperms' families, an interaction that is fundamental for biodiversity maintenance given that almost all plant species depend on it to complete their life cycle (Fleming & John Kress, 2011). Moreover, the role of bird frugivory affects not only seed dispersal, but also seed removal (Ruggera et al., 2016) and this role can be different among bird species within the same network. For example, bird species can be more or less important ('central') in a network depending on their interaction patterns with the available plant partners. Central species are usually generalists that feed on a large number of plant species without being limited to a particular group of plants; peripheral species are those with a higher degree of specialisation, which can be identified by their interaction with specific plant partners (Jordano et al., 2003; Olesen et al., 2007). Both roles, central and peripheral, are important for the maintenance of ecological networks, as generalists promote stability and functioning of the network (Guimarães Jr et al., 2011), whereas specialists yield unique ecosystem services (Mouillot et al., 2013).

But what determines bird species centrality in seed-dispersal networks? Some studies of local networks suggest that dietary specialisation is the most important driver (Escribano-Avila et al., 2018; Sebastián-González, 2017), as birds that consume fruits as their main diet component interact with a wider range of plant species and tend to occupy more central roles compared to frugivores that only consume fruits sporadically (Mello et al., 2015; Schleuning et al., 2011). Morphological traits, such as body mass (Pigot et al., 2016), have also been reported as relevant drivers of bird functional roles in plant–frugivore networks. On the one hand, more central species in these networks tend to have larger body size, which facilitate flexible frugivory choices (Jordano, 2017); on the other hand, small-size birds (lower body mass) are not capable of dispersing large seeds and tend to be more specialised in its interactions and occupy peripheral roles (Palacio et al., 2016). Still, the effect of body mass on species' centrality is not consistent across studies, since birds with smaller-to-medium body mass have also been found to occupy central roles in some seed-dispersal networks (Bastazini et al., 2019; Emer et al., 2018). These contrasting findings may be due to sampling bias (Acevedo-Quintero et al., 2020) or to the strong specialisation of some bird clades, leading to specific interactions between unique sets of plants and birds in a given network (Dehling et al., 2016; Muñoz et al., 2019).

Species' geographical distributions and climatic niches can also influence species' roles in interaction networks. For instance, species with larger range sizes tend to interact with more species, which would make them more central in a network. Conversely, species with narrower distributions tend to occupy peripheral positions (Dáttilo et al., 2020). Species' roles can also be influenced by regional effects, where structure of biotic interactions may be highly conserved in certain regions, regardless of species identity and environment (Muñoz et al., 2019; Vizentin-Bugoni et al., 2019). In addition, species interactions can also be constrained by their evolutionary history. In fact, species that are more central in seed-dispersal networks in tropical and temperate areas tend to be those coming from more evolutionary stable (low extinction and/or high speciation rates) lineages (Burin et al., 2021). The climatic niche breadth of species has also been postulated as a determinant factor to explain bird–plant specialisation, with birds having narrow climatic niches those who interact with a lower number of plants (Schleuning et al., 2016). However, most studies dealing with the drivers of species centrality have focused on the local network level (i.e. networks recorded at a specific place and time) thus neglecting the effect of such drivers on the role of species when considering all interactions across networks at the global extent.

Local seed-dispersal networks represent an opportunity to assess species contributions to ecosystem functions (Pigot et al., 2016). This is because they are representations of mutualistic interactions of local plant and bird assemblages that can also be connected to each other via individual species dispersion and colonisation dynamics (Hagen et al., 2012). Indeed, when ecological networks share species interactions, resulting in two or more networks to be connected across space (or time), they form a meta-network. Bird species that are repeatedly found in various local networks are those that perform more central roles, promoting the stability and cohesion of the meta-network structure (Martín González et al., 2010). The use of meta-networks has been applied to understand species' centrality in seed-dispersal networks in fragmented landscapes (Emer et al., 2018), to evaluate the effect of different trait characteristics in species centrality (Sebastián-González, 2017) and to assess how different interaction types are connected (Dáttilo et al., 2016). Still, whether the drivers of species centrality at the local network level are similar to those at the global meta-network level remains to be unveiled.

Here, by compiling a global dataset of 308 avian seed-dispersal networks across five continents and 11 biogeographical regions, we evaluated the relative importance of morphological, geographical, climatic, and evolutionary properties, in explaining species' centrality at the meta-network and local network levels. At the meta-network level, drivers acting at regional scales may determine species' centrality through their effect on their geographical distributions and realised climatic niches (Figure 1). That is, if a species is geographically common (e.g. with large range size or niche breadth), it would be expected to interact with a large set of plant species that are found in various local networks and therefore to be more important in structuring the meta-network, that is, there is a positive relationship between species' centrality and range size/niche breadth (Figure 1). Conversely, geographically/climatically restricted species, which are usually considered rare or specialists, may occupy more peripheral roles in the meta-network. At the local network level, we expected central species to be those that interact with a large set of plant species and thus increase the links between

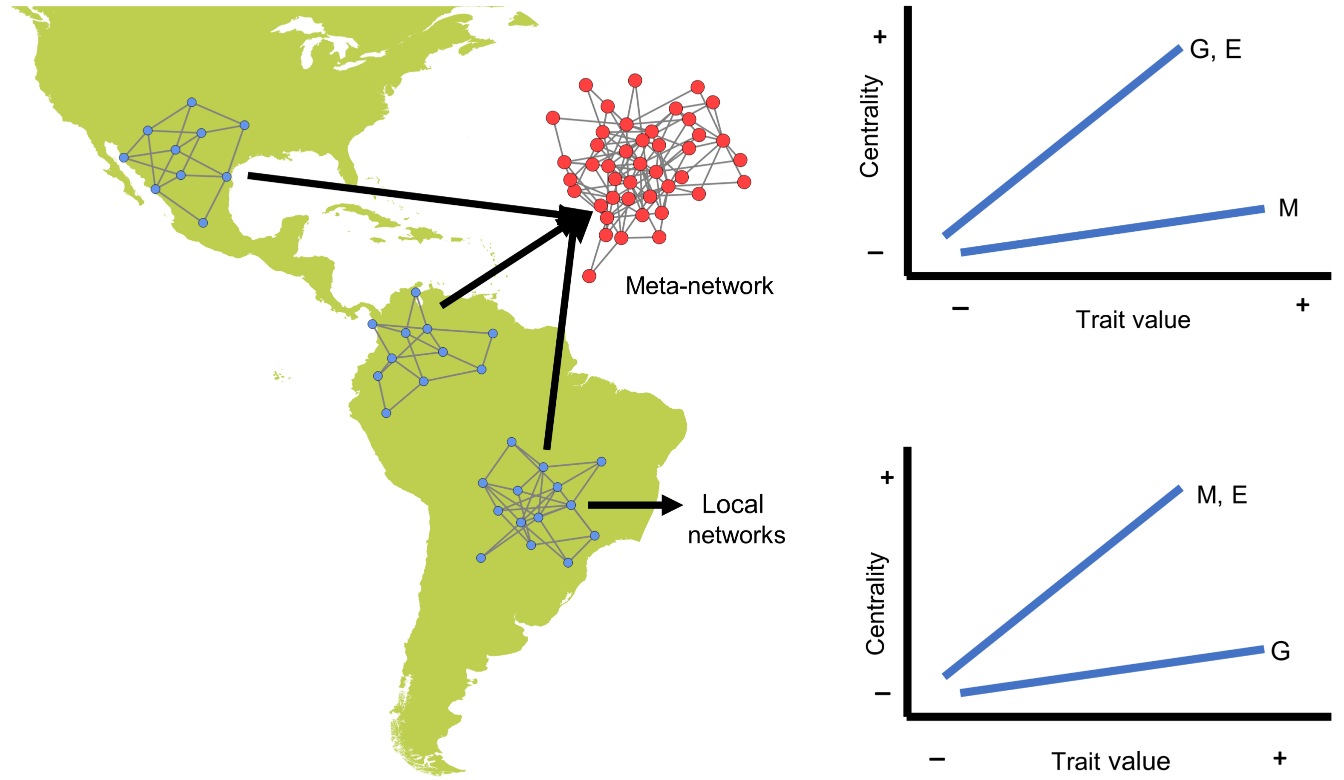


FIGURE 1 Schematic representation of the conceptual framework applied in this study. The map represents our broad spatial focus, with local networks representing individual observations of seed-dispersal networks and the meta-network being the combination of these local networks into a global network that considers all bird–plant interactions across local networks. The upper bivariate plot depicts the expected relationships between centrality and species' traits (G, geographical; E, evolutionary; M, morphological) at the meta-network level, with larger trait values (e.g. larger range size) being related to higher centrality. The lower plot depicts these relationships at the local network level. Note the expected difference in the magnitude of the traits' effect between the two network levels.

species. As such, geographical properties may be less influential in determining species' roles and drivers related to morphological traits may prevail. Regarding evolutionary properties, we expect that evolutionary stability determines species' centrality at both network levels given the influence of evolutionary dynamics on ecological and biogeographical patterns.

## 2 | MATERIALS AND METHODS

### 2.1 | Data gathering

Studies describing mutualistic networks of frugivory, or seed dispersal were compiled from the scientific literature, using the Web of Science (WoS), Scopus and Google Scholar databases. We used search strings that follow the word combinations: "seed dispersal network" OR "frugivory dispersal network" OR "plant-frugivory network" OR "mutualistic interaction network" AND "dispersal" in the title, abstract and keywords of the articles. In WoS truncated searches using '*' on those keywords were also applied to retrieve word variants. In addition, we searched for studies in two specialised databases: the Interaction Web Data Base (https://www.nceas.ucsb.edu/interactionweb) and the Web of Life (http://www.web-of-life.es/2.0/index.php). We also downloaded from GitHub (https://github.com) and DRYAD (https://datadryad.org) repositories published datasets. The flowchart of data compilation can be found in the Figure S1. Through this search, we compiled data from 308 bipartite networks from various sources until April 2022 from 11 biogeographical regions, as defined by (Holt et al., 2013; Figure S2).

In the compiled dataset, bird names were validated using the R package TAXISE (Chamberlain & Szöcs, 2013) and then harmonised using the taxonomic treatment of the Handbook of the Birds of the World and BirdLife International digital checklist of the birds of the world version 6 (HBW and Birdlife International 2021). We did not consider birds not identified to the species level. As for the plant names, they were mostly kept as originally downloaded from the datasets. Minor edits were made to adequate plant names when two or more entries were clearly the same (e.g. removal of the species author or the family name from the species identification). However, because we could not be sure if plant morphospecies (e.g. *Piper* sp.) present in different networks were unique entries in the datasets, we also prepared a subset of the original data considering only plants identified to the species level or those morphospecies that were clearly unique entries, which we then used for subsequent sensitivity analysis indicated below.

Bird morphological traits were obtained from the AVONET database (Tobias et al., 2022), namely Beak Length, Beak Width, Beak Depth, Tarsus Length, Wing Length, Kipps Distance, Hand-wing Index, Tail Length and Body Mass. Out of those traits, we selected a set of four morphological traits that would be informative of species' roles in the networks. These were body size, a proxy for species metabolism (Woodward et al., 2005); beak width, a proxy of frugivory (Dehling et al., 2014); hand-wing index (HWI), a trait associated with species dispersal (Sheard et al., 2020) and wing length, a trait associated both with species dispersal capacity (Dehling et al., 2014) and foraging behaviour (Bender et al., 2018).

Bird phylogenetic information was obtained by downloading 100 Phylogenetic trees—Stage 2 Hacket (Jetz et al., 2012) from the Global Phylogeny of Birds (www.birdtree.org). Because closely related species are expected to share similar traits, ecological redundancy is also expected to decrease with species evolutionary distinctiveness. So, we calculated the evolutionary distinctiveness (ED; typically quantified using phylogenetic branch lengths) for each species using the Equal Splits measure implemented in the function evol.distinct from the R package PICANTE (Kembel et al., 2010). We also calculated the DR rate (Jetz et al., 2012) for each species in each phylogenetic tree. DR is calculated as the inverse of ED using Equal Splits (1/ED) and often referred to as a metric of speciation rate (Title & Rabosky, 2019). Both ED and DR were calculated across 100 trees and the average value for each species was taken.

Birds' climatic niche requirements were calculated based on the birds' range maps obtained by request from BirdLife International (https://www.birdlife.org). From these range maps, we calculated for each bird species its geographical midpoint (mid Latitude and Longitude) and range size in square metres using the LETSR package (Vilela & Villalobos, 2015). The ecological niche of a species can be described by its mean position and breadth along environmental axes. Here, niche position and niche breadth along climatic axes were described using the multivariate analysis Outlying Mean Index (Dolédec et al., 2000). The climatic conditions of each species were characterised based on climatic variables downloaded from the WorldClim v2.1 (www.worldclim.org; Fick & Hijmans, 2017). Niche position represents a measure of the distance between the mean conditions used by the species and the mean conditions of the species group considered, which in our case was all birds. Niche breath is a metric that indicates the range of resources a species uses (Slatyer et al., 2013).

Using birds' distributional midpoints, we assigned each species to a biogeographical realm. Some species had their midpoints outside continental areas, to which we manually assigned the closest biogeographical realm. Birds' conservation status according to the IUCN RedList for the species detected in the networks was obtained from the classification of each species made available by (Tobias et al., 2022). Eighty-six per cent of the species of our dataset were classified in the category of 'Least Concern'. As such, we did not consider conservation status as a potential species property to evaluate its centrality.

## 2.2 | Data analysis

Each individual bipartite seed-dispersal network was considered as an independent dataset. Individual networks from the same location but observed in different years or seasons were also considered as independent datasets. Before data analysis, we excluded all species from the Psittacidae family (47 species) as their role as seed-dispersers is debated (Tella et al., 2015). In each network, interactions were transformed into binary interactions, that is, 0 (the absence of plant–bird species interaction) or 1 (the presence of plant–bird species interaction). The list of local individual networks was converted into a matrix (i.e. the meta-network) of interactions, with bird species in the columns and plant species in the rows. We made sure that all birds and plants had at least one interaction in the meta-network to avoid including smaller independent networks, and thus species, that are disconnected to the rest (Lever et al., 2014) and could thus bias the estimated role of individual species within the meta-network. For this, we removed all species that interacted with only one species from the other trophic level and were disconnected from the rest of the meta-network (only eight bird species were removed, 0.63%). The total number of bird species and plant species/morphospecies in our dataset after cleaning was 1264 and 2903 respectively. All analyses were done separately in two levels: at the meta-network and at the local networks individually. By performing analyses at these two network levels, we evaluated whether species are central across its range, that is, in each location the populations of such a species is also generalist (meta-network model); or whether species are a collection of populations specialised in different resources in each location (local network model).

Species' structural role in a network can be described by ranking species based on various indices of network centrality. Here we used species-level centrality metrics that describe different node (species) properties of both meta-network and each individual network, according to (Burin et al., 2021): (i) degree (the number of links of each species), (ii) betweenness (the proportion of network shortest paths the species participates in), (iii) closeness (shortest connections between a species and every other network species) and (iv) Katz centrality (a measure of the distance, in terms of all possible pathways, between the focal species and all other network species). These are commonly used centrality metrics that provide us with information about the importance of each species network structure when considering only the direct interactions (degree), the shortest pathways (closeness), the connecting pathways (betweenness) or all pathways (Katz; Burin et al., 2021; Oldham et al., 2019). With the exception of the Katz metric, which was calculated using a custom R code written by the second author, the other metrics were calculated using the R package BIPARTITE (Dormann, 2011). Therefore, following common practice in the ecological network literature (Maia et al., 2019; Medeiros et al., 2018; Sazima et al., 2010), the four metrics were combined using a principal component analysis (PCA) on a correlation matrix to obtain a single centrality metric that summarised the information on the interactive role of bird species at each meta-network and local network levels. At the meta-network level, PC1 explained 74% of the total variance and PC2 13%. At the local network level, the first and second PCs combined explained >90% of total variance for each network. Due to the high explained variance in both network levels, we used only the first principal component (PC1). Because the four-centrality metrics correlated positively with PC1 (Figure S3, see also Salavaty et al., 2020), the highest positive values were associated with those species that are more central in a network. For convenience when interpreting the PC1 values and ranking species from 0 (more peripheral) to the highest positive value (more central), PC1 values were rescaled to nonzero, positive-only values by adding up the absolute value of the minimum score plus a millesimal unit to each centrality value (Cruz et al., 2022). As such, all PC1 values were greater than zero while keeping the original order and distance among them (Dáttilo et al., 2022).

For the meta-network level, each metric was calculated for the whole meta-network and summarised using the PCA procedure. For the individual network level, each metric was first calculated for each network and standardised by calculating the z-scores to allow comparison among different networks. Then, standardised values were summarised using the PCA procedure. Because PC1 values were computed for each species in each individual network separately, and because bird species can occur in multiple networks, those species present in multiple networks had more than one centrality value (PC1). By doing that, we accounted for potential intraspecific variation, as the same species can be more central in one network and more peripheral in another. Forty-three individual networks from which the calculation of at least one of each of the four metrics returned non-numerical values, were excluded from the final analysis.

To assess the role of different trait variables in explaining species centrality, we applied Bayesian GLMMs (Table S1), which uses Markov chain Monte Carlo (MCMC) methods to obtain the models' likelihood to non-Gaussian data. We used the function MCMCglmm implemented in the R package MCMCglmm (Hadfield, 2010). For the meta-network, we designed two different models, in which we set morphological traits, evolutionary properties, niche characteristics and range size as fixed factors and species phylogenetic structure as a random factor (model 1), and a second model in which we added biogeographical realm affiliation as a second random factor (model 2; Table S1). For the local network model (model 3), the same fixed and random factors of model 2 were set (Table S1). For every model, each MCMC chain was run for 100,000 interactions with a burn-in of 25,000 and setting the model thinning parameter to 50. Noninformative priors were set for each residual structure (one for each model) and for every random term existent in the model, using the default values ($V=1$ and nu$=0.002$) defined in the MCMCglmm function. To account for phylogenetic uncertainty, we ran each model across 100 phylogenetic trees and evaluated the posterior means and posterior distribution of each factor. Model convergence was evaluated by visually inspecting trace plots of the fixed and random structures. All variables were standardised by subtracting the mean and dividing by the standard deviation prior to the analyses to allow comparison of effect sizes.

Then, we evaluated the effect of the different trait variables (fixed structure) in explaining species centrality (PC1) in each model by comparing the magnitude and significance of the posterior means and the 95% credibility intervals. The more posterior mean values deviated from zero, the stronger the effects. Also, we assessed variable significance by looking at the Particle Markov chain Monte Carlo (pMCMC), which is the proportion of coefficients in the posterior distribution estimates that are different from 0, multiplied by two (for a two-tailed test), which is analogous to a frequentist p-value (Penman et al., 2022). Regarding the random structure, we obtained the posterior means of each term and compared its magnitude between each other and if they deviated from zero, to assess their relative importance. Models' likelihood was assessed with the deviance information criterion (DIC), with the lowest value representing highest relative significance. DIC was calculated using the function DIC of the MuMIn package (Barton, 2018). Before running the models, we also checked for evidence of multicollinearity among predictors using variance inflation factors (VIFs) and found no evidence of severe (VIF > 10) or even moderate (VIF > 5) multi-collinearity (median VIF, 1.3; range, 1.03–4.985 for the meta-network model and median VIF, 1.54; range, 1.025–4.64).

We performed sensitivity analysis to assess if plant species that were considered as unique species but not identified to the species level could influence our results. To do so, we subsetted the original data set by removing those plant species not identified to the species level (or those that were unique and could not be mixed with another species although not identified at the species level) and their interaction with bird species. By doing that, the number of unique interactions decreased from 140,750 to 121,926 (~13% reduction) after removing the 652 species not identified to the species level (~22% of the original plant species number). Then, we performed the Bayesian GLMMs as with the original dataset and the resultant posterior means of all variables of the original and subset datasets were compared using ANOVA.

Finally, we tested whether the combination of the four-centrality metrics using PC1 correlated with a recently published metric, the Integrated Value of Influence (IVI), which is an integrative method for the identification of the most influential nodes within networks (Salavaty et al., 2020). The IVI is a combination of six metrics: degree, cluster rank, local H index, neighbourhood connectivity, betweenness and collective influence. IVI was calculated with the R package influential (Salavaty et al., 2020). IVI values were calculated for both meta-network and individual networks and used as dependent values in the Bayesian GLMMs (Table S1).

## 3 | RESULTS

### 3.1 | Characteristics of seed-dispersal networks

The bird species with higher frequency in the dataset were the common blackbird (*Turdus merula*, Turdidae) in 31% of networks, Eurasian blackcap (*Sylvia atricapilla*, Sylviidae) and song thrush

(*Turdus philomelos*, Turdidae) in ca. 25% of networks, the European robin (*Erithacus rubecula*, Muscicapidae) and the garden warbler (*Sylvia borin*, Sylviidae) in ca. 20% of networks. About 10% of the bird species were present in at least 10 different networks; 4% of the species were present in a single network only and 60% of the species were detected in more than one network.

The ranking of the 10 more central species, that is, those with higher PC1 values, included the Sayaca tanager (*Thraupis sayaca*, Thraupidae), rufous-bellied thrush (*Turdus rufiventris*, Turdidae), great kiskadee (*Pitangus sulphuratus*, Tyrannidae), blue dacnis (*Dacnis cayana*, Thraupidae), russet-backed thrush (*Catharus ustulatus*, Turdidae), pale-breasted thrush (*Turdus leucomelas*, Turdidae), palm tanager (*Thraupis palmarum*, Thraupidae), white-necked thrush (*Turdus albicollis*, Turdidae), red-eyed vireo (*Vireo olivaceus*, Vireonidae) and the Eurasian blackcap (*Sylvia atricapilla*, Sylviidae). Except for *C. ustulatus*, *V. olivaceus* and *S. atricapilla*, all other more central species are Neotropical. The number of distinct pairwise interactions in the meta-network was 140,750, with a mean number of links per species (degree) of 14.2.

For the set of local networks, the more central species were the emerald toucanet (*Aulacorhynchus* prasinus, Ramphastidae), grey-cheeked fulvetta (*Alcippe morrisonia*, Alcippeidae), Eurasian blackcap (*Sylvia atricapilla*, Sylviidae), blue-banded toucanet (*Aulacorhynchus coeruleicinctis*, Ramphastidae), white-necked thrush (*Turdus albicollis*, Turdidae), Cape white-eye (*Zosterops pallidus*, Zosteropidae), red-necked tanager (*Tangara cyanocephala*, Thraupidae), and the pale-breasted thrush (*Turdus leucomelas*, Turdidae). Except for *A. morrisonia* (Sino-Japanese), *S. atricapilla* (Palearctic) and *Z. pallidus* (Afrotropical), the rest of the species are distributed in the Neotropical region.

## 3.2 | Predictors of species centrality in the meta-network

The correlation between drivers can be found in Figure S4. At the meta-network (model 1), range size constantly deviated from zero in all the 100 runs (pMCMC range < 0.05, posterior mean distribution = 0.44, Figures 2a and 3). Morphological traits such as beak width (positive association, posterior mean distribution = 0.16) as well as HWI and body mass (both with negative association) also had posterior distributions that deviated from zero (posterior means of −0.11 and −0.08 respectively), but none of them had pMCMC range values below the significance threshold of 0.05 (Figure S5). Variables describing climatic niches and evolutionary properties such as niche position, niche breadth, evolutionary distinctiveness (ED) and diversification rate (DR) as well as another morphological trait, wing length, had almost no deviation from zero nor pMCMC range values below the significance threshold of 0.05 (Figure 2a, Table 1). Including species' affiliation to a biogeographical realm as a random term in a second meta-network model (model 2, Figure S6) produced similar results. Range size positively deviated from zero (pMCMC range < 0.05, posterior mean distribution = 0.50, Figures 2b and 3). Beak width (positive association) and body mass (negative association) also deviated from zero (posterior means of =−0.15 and −0.08 respectively), whereas the other trait variables had minimum

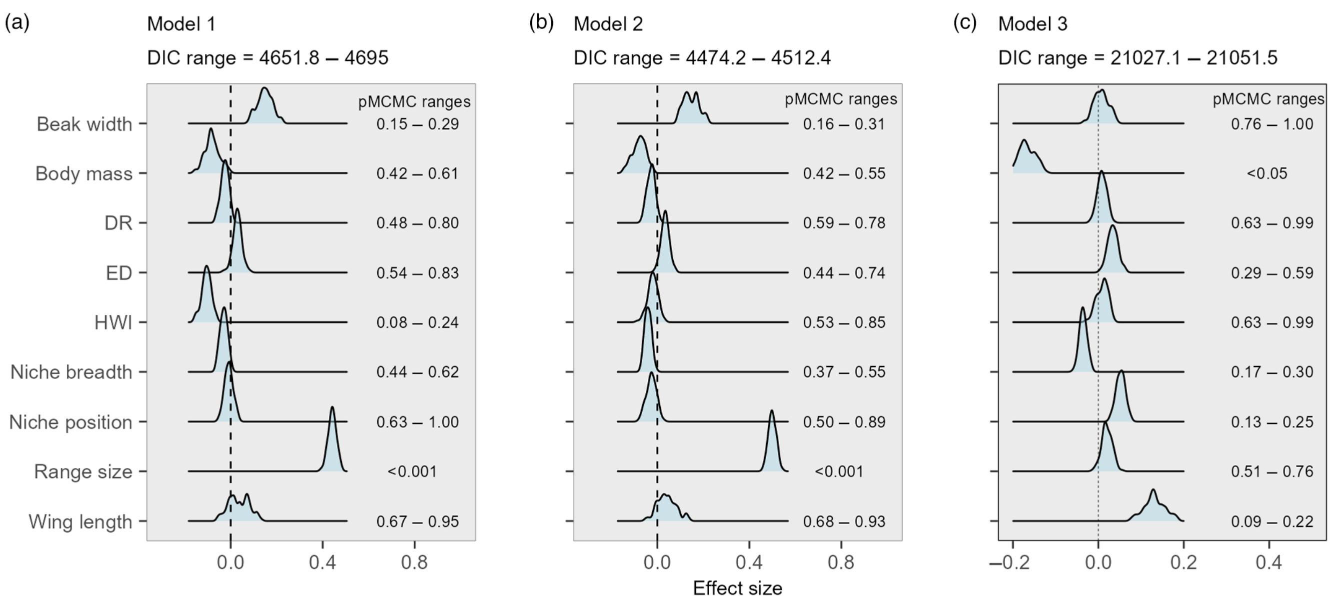


**FIGURE 2** Density distribution of posterior means of each predictor trait (in light blue) as calculated for 100 phylogenetic trees in each model (Table S1). Distributions with higher deviation from zero represent larger effects of the associated predictor trait in explaining species centrality (PC1). Graphics subtitles indicate the range values of the deviance information criterion (DIC) for each model as calculated for 100 phylogenetic trees. The range of pMCMC values, an analogous of frequentist *p*-values, for each model are indicated on the right side of each trait's density distribution. Models 1 and 2 were fit at the meta-network level, but Model 2 included species' biogeographical affiliation as a second random term in addition to the phylogenetic effect. Model 3 was fit at the local network level, considering all networks as individual observations. ED, Evolutionary distinctiveness; HWI, hand-wing index. Model's structure can be found in the methods (Table S1).

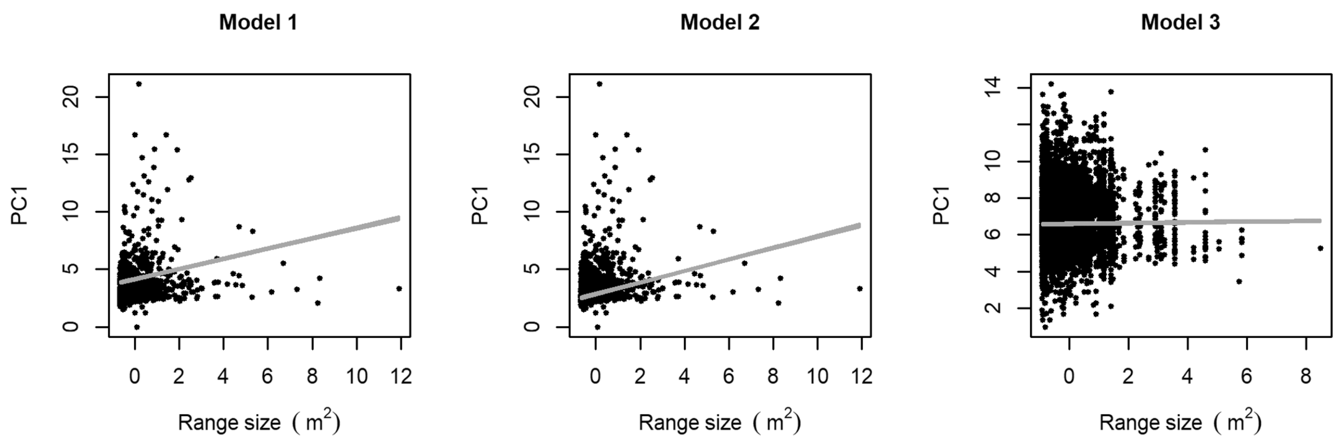


**FIGURE 3** Relationship between range size and species centrality (PC1) for models 1, 2 and 3. Raw data are shown with overlapping adjusted regression lines calculated with the average slope and intercept values obtained from all models using the 100 phylogenetic trees. Only for models 1 and 2, pMCMC values of range size were below the significance threshold of 0.05. The complete relationships between centrality (PC1) and bird traits for each model can be found in the Supplementary Data. The first and second PCs in combination explained 75% of total variance for the meta-network. In the local networks, the first and second PCs overall explained >90% of total variance. Range size was standardised prior to analysis.

**TABLE 1** Bayesian phylogenetic mixed model results for the effect of the predictor variables on species centrality in the three evaluated models (Table S1). All variables were standardised (mean = 0, SD = 1) prior to model fitting. All models were run over 100 phylogenetic trees. DR, Diversification rate; ED, Evolutionary distinctiveness; HWI, Hand-wing index.

| | Estimate (95% credible interval) | | |
|---|---|---|---|
| Variable | Model 1 | Model 2 | Model 3 |
| Beak width | 0.16 (−0.09 to 0.41) | 0.15 (−0.09 to 0.38) | 0.01 (−0.14 to 0.15) |
| Body mass | −0.08 (−0.32 to 0.15) | −0.08 (−0.31 to 0.15) | −0.17 (−0.3 to −0.04) |
| ED | 0.03 (−0.1 to 0.16) | 0.04 (−0.09 to 0.17) | 0.03 (−0.05 to 0.11) |
| DR | −0.03 (−0.14 to 0.09) | −0.03 (−0.14 to 0.09) | 0.01 (−0.06 to 0.08) |
| HWI | −0.11 (−0.26 to 0.04) | −0.03 (−0.17 to 0.12) | 0.01 (−0.08 to 0.1) |
| Niche breadth | −0.03 (−0.13 to 0.07) | −0.04 (−0.14 to 0.06) | −0.04 (−0.09 to 0.02) |
| Niche position | −0.01 (−0.12 to 0.11) | −0.03 (−0.17 to 0.11) | 0.05 (−0.03 to 0.13) |
| Range size | 0.44 (0.33 to 0.55) | 0.5 (0.38 to 0.62) | 0.02 (−0.06 to 0.1) |
| Wing length | 0.04 (−0.26 to 0.34) | 0.04 (−0.25 to 0.33) | 0.13 (−0.05 to 0.3) |

deviations and none of them had pMCMC range values below the significance threshold of 0.05 (Figure 2b, Table 1). Credible intervals of posterior mean distributions can be found in Table 1.

Model 2 had smaller deviance information criterion (DIC) values than Model 1, meaning that the inclusion of the random term biogeographical realm improved the model's performance (Figure 2a,b). However, when comparing the effect of both random terms, the phylogenetic structure was more important than the biogeographical realm in all models regardless of network level, as its posterior means had higher deviations from zero, ranging from 0.69 to 1.49, whereas the posterior means of the biogeographical realm had lower values ranging from 0.27 to 0.32.

### 3.3 | Predictors of species centrality in local networks

Contrary to the meta-network, at the local network level (Model 3) range size of bird species was not associated with their centrality (posterior mean distribution = 0.02, Figures 2c and 3). Instead, morphological traits such as body mass (negative association, pMCMC <0.05) and wing length (positive association, pMCMC range 0.09–0.22) were those whose posterior distributions had larger deviations from zero (posterior mean of −0.17 and 0.13 respectively) compared to the other drivers (Figure 2c, Table 1, Figure S7). In this model 3, the biogeographical realm had a small relevance (posterior means ranged from 0.005 to 0.006), whereas the phylogenetic structure was predominantly relevant (posterior means ranged from 0.88 to 0.90). Credible intervals of posterior mean distributions can be found in Table 1.

The sensitivity analysis indicated that for model 1, DIC range values decreased from 4651–4695 to 4536–4555; for model 2, from 4474–4512 to 4366–4381; for model 3, from 18,876–18,877 to 17,986–18,003. Similar posteriori means were found between comparing the two datasets (Figure S8) and no significant difference mean differences were detected (Table S2). Regarding the comparison between the IVI and the PC1, both indices were correlated when measured at the meta-network level (Pearson's $r = 0.64$, $p < 0.05$)

producing very similar results regarding predictor variables of centrality at this level (Figure S9), but not so much for the individual networks (Pearson's $r=0.01$, $p=0.77$; Figure S9) perhaps given their high variation in terms of size (number of species) and connectance as well as the different number of variables used in both compound metrics (four in PC1 vs. six in IVI).

## 4 | DISCUSSION

Our results revealed that the factors that govern species' centrality were not the same at the meta-network and at the local network levels. According to our predictions, geographical traits were more important than morphological traits or evolutionary properties in explaining species' centrality at the meta-network but not at local networks. Specifically, we found that the more central species were those with larger range sizes at the meta-network level, whereas at the local networks species' range sizes had no significant effect in explaining species centrality. Instead, our results revealed that morphological traits such as body mass was the most important drivers of centrality at the local individual networks. This difference in the drivers of species' roles and their importance indicates that the process governing species centrality is dependent on the network level.

Bird species that occupy more central roles at the meta-network were those that have larger range sizes. Large range size increases the chance of a species to interact with a larger set of partners (e.g. plants) across its geographical distribution (Dáttilo et al., 2020). Range size correlates with dispersal capacity of bird species (Arango et al., 2022), so it might also be possible that long distance travel ability fosters the species' role as central by keeping the links in the meta-network. Moreover, central species may maintain meta-network links by feeding in different regions via flexible diets over seasons. However, species dispersal may be restricted across space because of biogeographical barriers or by landscape fragmentation. Still, central species can maintain the cohesion of the meta-networks even in fragmented landscapes. In these cases, those central species have been reported to be generalist small-bodied or large-bodied birds (Lenz et al., 2011). Both can disperse seeds across fragmented landscapes and thus help maintaining meta-network cohesion regardless of their size and/or centrality. As such, the influence of dispersal capacity in determining species centrality within meta-networks requires detailed evaluation that considers other species' traits to disentangle their combined effect on the structure of this biotic interaction.

At the local network level, our expectation was that morphological traits would be more relevant in driving species' centrality. In this study, bird's body mass was the trait that had more association with species centrality. The observed negative association of body mass suggests that peripheral species would be those with larger body mass, which agrees with a previous study in fragmented areas, implying that large-bodied species are less central in seed dispersal at small spatial scales (Emer et al., 2018), or that large-bodied species may be peripheral in networks from fragmented landscapes given their low abundances in such contexts, where they are more subject to hunting (Vidal et al., 2013). Our findings showed that the other traits selected (i.e. beak width, wing length and HWI) had neither positive (explained central roles) nor negative effects (explained peripheral roles) on species' centrality. The absence of a strong effect from these selected traits could be related to their lack of relevance in determining species centrality in local networks or because our considered traits are species level and thus do not reflect intraspecific variation, which could indeed affect the interactive role of the individuals of bird species at the particular locations where they participate in bird–plant interactions. Alternatively, our independent variables could be related to PCA axes (besides the PC1) that had explained small part of the total variance, and therefore were not used in our analysis. Moreover, other ecological aspects can help elucidating the drivers of species' functions at individual networks, such as species abundances. For instance, for other fruit dispersers such as bats, species relative abundances were more determinant than morphological traits in explaining species' ecological roles (Laurindo et al., 2020).

A recent study indicated that birds that occupy more central roles at local networks are also those that belong to lineages with higher macroevolutionary stability through low extinction and/or high speciation (Burin et al., 2021). These authors suggested an interplay between interaction patterns at local and contemporary scales and species diversification at broader spatial extents and temporal scales. In our analysis, none of the evaluated evolutionary properties (evolutionary distinctiveness and tip diversification rate) were significant in explaining species centrality at the local network nor at the meta-network level. Similarly, another study on the relative effect of different traits on species' roles did not detect evolutionary signals in seed-dispersal networks (Pigot et al., 2016). The main difference between these studies and ours is that we considered more local networks (308 vs. 29 and 35 in Burin et al. (2021) and Pigot et al. (2016) respectively) and thus a broader spatial coverage. As such, our finding of species' centrality being unrelated to their evolutionary properties could be explained by the mismatch between processes acting at different spatial–temporal scales in determining species' interaction, with species' roles in interaction networks perhaps being determined by ecological factors (not identified in our study) instead of evolutionary factors. Alternatively, the consideration of all bird species reported to be seed dispersers aggregates several different clades of birds that may have distinct evolutionary dynamics, thus obscuring the effect of species' evolutionary properties on their interaction patterns. In addition, the absence of such evolutionary effect could be due to the fact that our metrics (ED and DR) did not consider assemblage structure, as this kind of information is rarely available at broad spatial extents such as ours (but see Callaghan et al., 2021). Thus, the application of the macroevolutionary approach and the consideration of particular characteristics of individual networks, such as species local abundances and the frequency of interactions (Peralta et al., 2020), might help to better disentangle the role of lineage stability in explaining species centrality at local and meta-network levels in further studies.

The most central birds in our dataset were species from the families Thraupidae and Turdidae. Birds from these families have been considered as keystone species (Escribano-Avila et al., 2018). In addition, these central species tend to be tolerant of human presence and have flexible diets (Winkler et al., 2020a, 2020b). Besides, they are easily detected and identified because they are more conspicuous (Baldiviezo et al., 2019). A previous study raised the concern of common species becoming widespread across mutualistic networks (Fricke & Svenning, 2020), mainly because disturbance pressure may cause generalist species to become those with more central roles, and such generalists being, quite often, introduced non-native species. Common generalist species tend to interact disproportionately with other species even though they might disperse less seeds (Pejchar, 2015). All these processes are likely to amplify biotic homogenization and may reduce ecosystem resilience by exposing different ecosystems to similar drivers and species central roles may be related to the capacity of a species to occupy more available niches (Sales et al., 2021; Vidal et al., 2014).

We are aware that despite having sampled networks from five continents and 11 biogeographical regions, our dataset is biased towards areas with more data coverage that could limit our conclusions. The Neotropical region accounts for about half of our compiled networks, and continental areas are better covered than islands or coastal areas. It also known that species within the same biogeographical region tend to interact more and that strong species turnover between seed-dispersal networks occurs between biogeographical regions (Martins et al., 2022). Moreover, within the same biogeographical region, birds with medium to large body size or flexible diets tend to have greater range sizes due to the ability of crossing potential dispersal barriers (Claramunt et al., 2022). The addition of biogeographical affiliation indeed improved our meta-network models, indicating that this is an important source of variation of species centrality. Overall, despite the inevitable geographical bias in our compiled dataset, we were able to identify some drivers of bird species centrality in seed-dispersal networks and that these depend on the network level considered.

The traits used in our analyses were defined based on their reported association with species centrality in previous studies (Burin et al., 2021; Pigot et al., 2016). Because a single trait value is associated with each species, there may be a mismatch in the scale of our dependent and independent variables when considering the traits at local network level, as the selected traits do not consider intraspecific variation. Still, using species-level values rather than population-level trait values (which in fact are not available for most of the individual networks in our study), allowed us to consider other attributes such as geographical and evolutionary ones that emerge at the level of the species instead of the individual/population level.

Bird species are facing a long-term decline due to habitat loss and fragmentation (Pollock et al., 2022). Several species are being replaced by invasive non-native species that may not necessarily replace ecological function. By identifying the drivers of bird species centrality, we found that those species that currently occupy these central roles at the meta-network level are those species that have larger range sizes. Accordingly, habitat fragmentation may limit species movements and migrations affecting the structure of interaction networks as well as the ecosystem function of different species. At local scales, other studies have shown that central frugivore species can also be those under higher extinction risk, mainly given their vulnerability to habitat loss and hunting (Vidal et al., 2014). Moreover, climate change has also been identified as a potential driver of network restructuring, affecting the mutualism between seed dispersers and plants (Sales et al., 2021; Schleuning et al., 2016) and colonisation-extinction dynamics (Sonne et al., 2022) as both interacting groups may rearrange to track suitable climatic conditions. Thus, we were able to show that bird ecosystem functions are determined by species capacity to occupy various habitats and interact with a large set of plant species. Still, the interaction links between plants and birds are dynamic and may change over time. In fact, the importance of species-level traits in determining species centrality in temporally dynamic networks deserves further attention. This is important to understand how species interaction roles will be affected under the increasing anthropogenic impacts on biodiversity.

## AUTHOR CONTRIBUTIONS

Gabriel M. Moulatlet, Wesley Dáttilo, Fabricio Villalobos conceived the ideas, designed methodology and collected the data; Fabricio Villalobos secured funding; Gabriel M. Moulatlet analysed the data and led the writing of the manuscript with help from Fabricio Villalobos. All authors contributed critically to the drafts and gave final approval for publication.

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## CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

## DATA AVAILABILITY STATEMENT

Data available from the FigShare Digital Repository https://figshare.com/s/6d409efbed4f9d0c239e (Moulatlet et al., 2023).

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## SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

**Figure S1.** Flowchart of data compilation and analysis.

**Figure S2.** Distribution of the seed-dispersal networks used in this study over the biogeographic realms.

**Figure S3.** Pairwise correlations between centrality metrics as calculated for (A) the meta-network and (B) the individual networks.

**Figure S4.** Pairwise Pearson's correlation between the selected drivers of species centrality used in the modelling.

**Figure S5.** Relationship between species centrality (PC1) and each of the evaluated traits for model 1. The relationship between PC1 and the variable range size is depicted in Figure 3.

**Figure S6.** Relationship between species centrality (PC1) and each of the evaluated traits for model 2. The relationship between PC1 and the variable range size is depicted in Figure 3.

**Figure S7.** Relationship between species centrality (PC1) and each of the evaluated traits for model 3. The relationship between PC1 and the variable range size is depicted in Figure 3.

**Figure S8.** Comparison of density distribution of posterior means of each predictor trait for the original dataset and for a data subset including those plant species that have been identified to the species level.

**Figure S9.** Comparison of density distribution of posterior means of each predictor trait as having the PC1 and the IVI as predictors of species centrality (dependent variables).

**Table S1.** Fixed and random structures of MCMCglmm models.

**Table S2.** Differences between the posterior means as obtained with comparison of the original dataset and with the subset of the original data that included only plant species identified to the species level using Analysis of Variances (ANOVA).

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